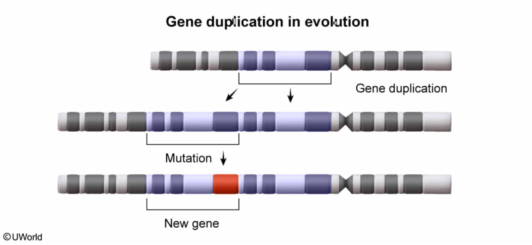
A certain eukaryotic organism has multiple, distinct genes with highly similar sequences that are expressed at different times in the organism's life cycle. These genes could have arisen by:

- A. alternative splicing. [59%]
- B. conjugation. [5%]
- C. gene duplication. [29%]
- D. transformation. [5%]

Correct
29% answered correctly

Explanation:



Genes with similar sequences are considered **evolutionarily related**. In other words, they are believed to have a common origin in a single gene from an ancestor organism. During the process of molecular evolution, portions of a chromosome can be **duplicated** by mechanisms such as unequal crossing over. Over time, duplicate genes can undergo mutations, causing them to have similar but not identical sequences. These differences allow the genes to carry out **distinct roles** at various times in an organism's life cycle.

The question states that some organisms have multiple genes with highly similar sequences. These similar genes most likely arose by **gene duplication**.

(Choice A) Alternative splicing produces multiple protein products by producing distinct mRNA molecules from the *same gene* (ie, the same segment of DNA). In contrast, the question asks about *distinct genes* (ie, distinct segments of DNA) with similar sequences.

(Choice B) Conjugation is the exchange of genetic information between prokaryotes, typically in the form of plasmid DNA. It does not occur in eukaryotes.

(Choice D) Transformation occurs when prokaryotes pick up foreign genetic material from their surroundings. Eukaryotes do not participate in this process of acquiring genetic material.

Educational objective:
Distinct genes within an organism that have high sequence identity are evolutionarily related, likely having a common origin. They generally arise by gene duplication and, over time, can mutate and fulfill distinct roles within an organism.

Time spent:0 sec
QID:403707
Subject: Biology
System: Molecular Biology

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Each time a eukaryotic cell replicates its DNA, the telomeres at the ends of each chromosome become slightly shorter. When telomeres reach a critically short length, the cells enter a senescent state, in which cell division ceases (Figure 1). In humans, conditions such as hypertension, liver disease, and Alzheimer disease have been associated with the decreased telomere length and resulting senescence that occur as part of the normal aging process.

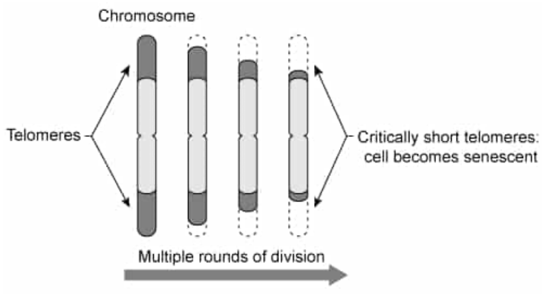

Figure 1

Cells that must be able to divide indefinitely (stem cells and germ cells) express an enzyme known as telomerase. Functional telomerase consists of a protein subunit called telomerase reverse transcriptase (TERT) and a noncoding RNA subunit called telomerase RNA (TR). Using a portion of TR as a template, telomerase extends telomeres by continuously adding the sequence 5′-TTAGGG-3′ to the ends of chromosomes. The complement strand is then filled in by DNA polymerase. Telomere extension allows these cells to avoid senescence.

However, in healthy somatic cells telomerase expression is suppressed, and the resulting critical telomere shortening has been hypothesized to prevent uncontrolled cell proliferation in adult tissues. Increased telomerase activity is often associated with tumorigenesis. *TERT* transcription is generally controlled by chromatin structure as well as the oncogene c-Myc and the tumor suppressor protein WT1. Healthy cells express relatively high levels of WT1 and low levels of c-Myc whereas cancerous cells often express low levels of WT1 and high levels of c-Myc, leading to increased TERT expression.

In addition to transcriptional control, TERT can be regulated at the post-transcriptional level. The *TERT* gene consists of 16 exons and 15 introns that can be spliced into more than 20 different isoforms in humans. Only one isoform (the active isoform) is known to extend telomeres. However, the i2 isoform, in which intron 2 is partially included in the mature mRNA (Figure 2), has been hypothesized to downregulate residual TERT activity in somatic cells.

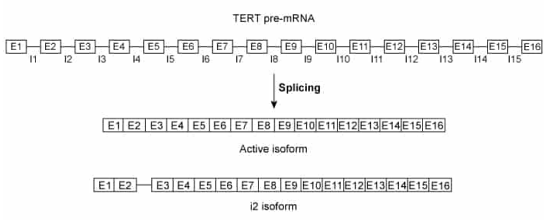

Figure 2

WT1 and c-Myc most likely alter transcription levels of the gene that encodes TERT by doing which of the following at the *TERT* gene promoter?

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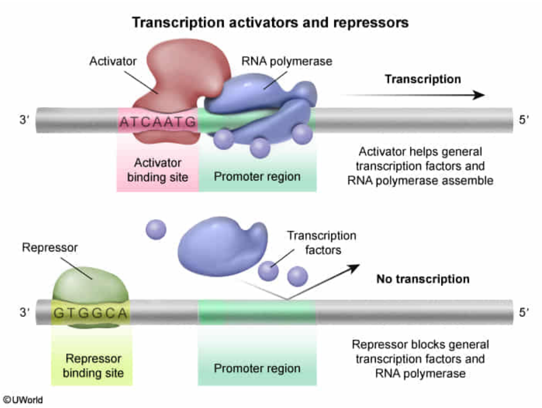
- [x] A. Inhibiting and facilitating RNA polymerase binding, respectively [48%]
- [ ] B. Inhibiting and facilitating DNA polymerase binding, respectively [29%]
- [ ] C. Facilitating and inhibiting RNA polymerase binding, respectively [12%]
- [ ] D. Facilitating and inhibiting DNA polymerase binding, respectively [8%]

Correct
49% answered correctly

Explanation:



RNA polymerases are responsible for carrying out transcription and must **bind DNA at a promoter** region for transcription to be initiated. **Transcription factors** are proteins that can bind to DNA near gene promoter regions and either **increase** transcription **(activators)** or **decrease** transcription **(repressors)**. **Activators facilitate** RNA polymerase binding to the promoter, and **repressors inhibit** binding.

According to the passage, WT1 is present in cells that do *not* express *TERT* (eg, healthy somatic cells) and absent in cells that *do* express it (eg, cancer cells, somatic cells, germ cells). The opposite occurs with c-Myc, which is absent in cells that do *not* express *TERT* and present in cells that *do*.

The presence of WT1 in healthy somatic cells, where the *TERT* gene is transcribed at low levels, suggests that WT1 inhibits *TERT* expression in these cells at the transcriptional level. In contrast, c-Myc most likely *promotes TERT* transcription. Based on this information, WT1 likely acts as a repressor and c-Myc likely acts as an activator. Therefore, WT1 and c-Myc must **inhibit and facilitate RNA polymerase binding, respectively**.

(Choices B and D) DNA polymerase is involved in replication, not transcription, so it is not involved in the transcription of the *TERT* gene. Transcription factors do not affect DNA polymerase binding to DNA.

(Choice C) WT1 is a repressor whereas c-Myc is an activator. WT1 *inhibits* binding and c-Myc *facilitates* binding.

Educational objective:
Transcription factors can upregulate or downregulate transcription by influencing the ability of RNA polymerase to bind a promoter. Transcription factors that increase transcription are called activators and facilitate RNA polymerase binding whereas those that decrease transcription are called repressors and inhibit binding.

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Time spent:0 sec
Subject:
Biology

QID:403708
System:
Molecular Biology

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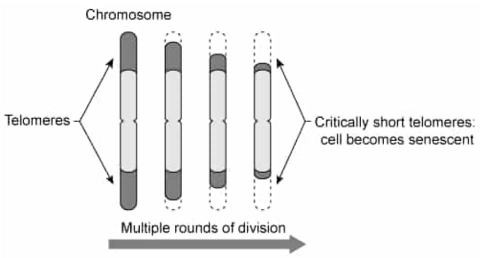

Figure 1

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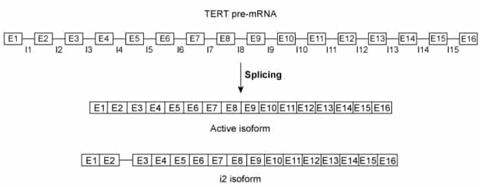

Figure 2

Given that in healthy somatic cells the *TERT* gene is rarely transcribed, where in the chromosomes of these cells is the gene most likely found?

- A. In a relatively open, uncoiled portion of the chromosome [5%]
- B. In the portion of the chromosome that binds the kinetochore [6%]
- C. In a portion of the chromosome that associates with ribosomes [2%]
- ✔ D. In a portion of the chromosome that is tightly wound around histones [85%]

Correct
86% answered correctly

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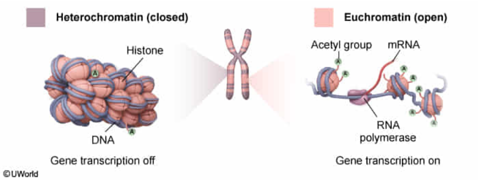
Explanation:



Gene transcription from DNA to RNA depends partly on the **chromatin structure** of the DNA that contains the gene. Broadly speaking, chromatin exists in two forms: heterochromatin and euchromatin. **Heterochromatin** consists of DNA that is **tightly coiled around histone proteins**, bound by an ionic interaction between the negatively charged phosphates on the DNA backbone and positively charged lysine residues in the histone. DNA in heterochromatin is **not readily accessible** to RNA polymerase and so **cannot be readily transcribed**.

Euchromatin forms when histones are modified, often by acetylation of lysine residues. The added acetyl group neutralizes the positive charge on a lysine residue, reducing interactions between histones and DNA. The reduced interactions yield a **more open form** that is **more accessible** to RNA polymerase, allowing euchromatin to be more **readily transcribed**.

Because somatic cells rarely transcribe *TERT*, the gene that encodes TERT is most likely found in a heterochromatin form most of the time within these cells. Therefore, in somatic cells the TERT gene is most likely in a portion of the chromosome that is **tightly wound around histones**.

(Choice A) Relatively open, uncoiled DNA is characteristic of *euchromatin*. As indicated in the passage, stem cells and germ cells express the TERT gene at high levels, so the gene is likely to be found in euchromatin. However, this is not the case in somatic cells, which rarely express TERT.

(Choice B) The portion of the chromosome that interacts with the kinetochore during mitosis is the centromere. Although the centromere contains heterochromatin, the *TERT* gene is unlikely to be in this region because the DNA of centromeres typically encodes few genes.

(Choice C) Chromosomes are found in the nucleus and do not associate with ribosomes, which are found in the cytosol and on the rough endoplasmic reticulum. It is mRNA, not chromosomes, that associates with ribosomes.

Educational objective:

Chromatin can be broadly classified as heterochromatin or euchromatin. Heterochromatin consists of DNA tightly coiled around histones and is not readily transcribed by RNA polymerase. Euchromatin is more loosely associated with histones and is more easily transcribed.

Time spent:0 sec	**QID**:403709
Subject:	**System:**
Biology	Molecular Biology

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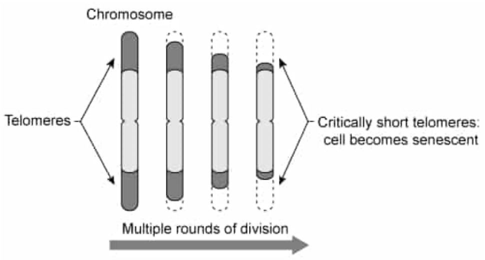

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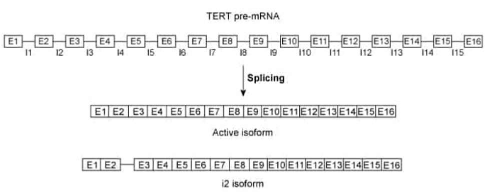

Figure 2

Based on the sequence that telomerase adds to the ends of chromosomes, the template portion of TR most likely contains which of the following sequences?

- A. 5′-UUAGGG-3′ [5%]
- ✔ B. 3′-AAUCCC-5′ [61%]
- C. 5′-CCCTAA-3′ [21%]
- D. 3′-CCCUAA-5′ [11%]

Correct

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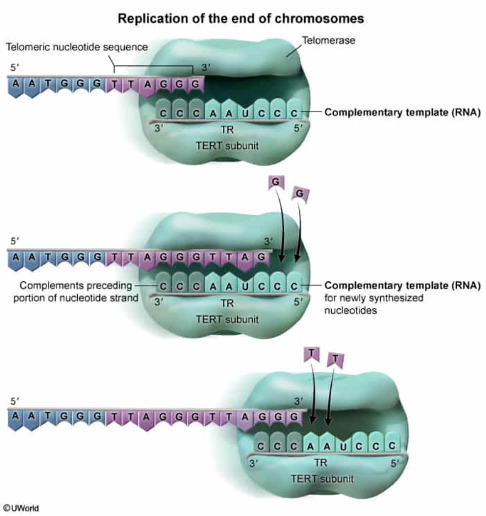
62% answered correctly

Explanation:



In DNA synthesis, the **template** strand is **always complementary** to the new strand that is being synthesized. Complementary nucleotide strands always align antiparallel to each other (ie, the 5′ end of one strand aligns with the 3′ end of the other). When the template strand is a DNA molecule, guanine (G) is always paired with cytosine (C) and adenine (A) is always paired with thymine (T). However, when the template strand is an RNA molecule, it contains uracil (U) instead of thymine.

The passage states that TR is the RNA subunit of telomerase that serves as a template for the TERT subunit to lengthen telomeres. Therefore, TR must contain uracil at any position where adenine is to be added to the telomere. In addition, because telomerase extends telomeres by continuously adding the sequence 5′-TTAGGG-3′ to the ends of chromosomes, the **TR subunit** must **contain 3′-AAUCCC-5′** as the **complementary template** sequence.

(Choice A) 5′-UUAGGG-3′ is the RNA equivalent of the sequence added to the end of telomeres (5′-TTAGGG-3′), with thymine (T) changed to uracil (U). It is not the complement of 5′-TTAGGG-3′.

(Choice C) 5′-CCCTAA-3′, which can also be written as 3′-AATCCC-5′, is a DNA strand that is complementary to the telomere sequence 5′-TTAGGG-3′. However, because the template component of telomerase is an *RNA* strand, thymine (T) should be changed to uracil (U) to produce the correct sequence.

(Choice D) 3′-CCCUAA-5′ is the reverse of the correct sequence (3′-AAUCCC-5′). This sequence, which can also be written as 5′-AAUCCC-3′, would have to align *parallel* to the 5′-TTAGGG-3′ sequence for complementary bases to align. However, the geometry of parallel strands prevents complementary bases from pairing with each other.

Educational objective:

New nucleotide strands are synthesized using complementary strands as templates, with the two strands aligned antiparallel to each other (ie, the 5′ end of one strand aligns with the 3′ end of the other). In DNA, guanine (G) always pairs with cytosine (C), and adenine (A) always pairs with thymine (T). In RNA, the only difference is that uracil (U) is present instead of thymine.

Time spent: 0 sec
Subject: Biology
QID: 403710
System: Molecular Biology

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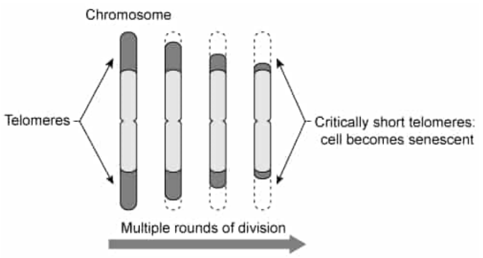

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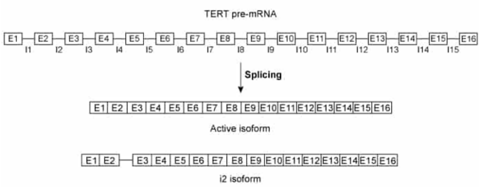

Figure 2

If the role of the i2 isoform of the *TERT* gene proposed in the passage is correct, individuals who express the i2 isoform in somatic cells would be expected to have a decreased risk of:

- A. hypertension. [1%]
- ✔ B. cancer development. [94%]
- C. Alzheimer disease. [3%]
- D. liver disease.

Correct
95% answered correctly

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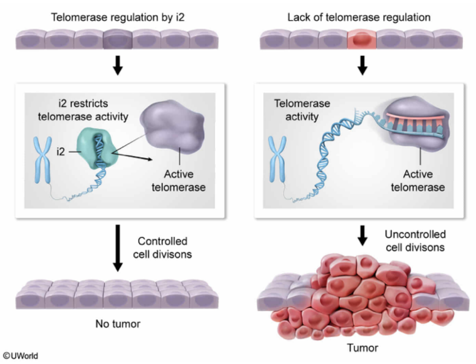
Explanation:



When protein-encoding DNA is first transcribed to RNA, it is in an immature form known as precursor mRNA (pre-mRNA). Pre-mRNA contains both **introns and exons** and is spliced into one of several possible mature mRNA forms, known as **isoforms**. Each isoform contains a subset of the exons in the pre-mRNA and may also contain portions of introns. Different mRNA isoforms lead to gene products (proteins) with different properties and are often used to **regulate gene activity**. For example, splicing pre-mRNA into an inactive isoform dilutes the amount of active isoform present, and the inactive form can compete with the active form for substrate, reducing the overall activity. Such downregulation might be evolutionarily advantageous as it decreases the harmful effects of an overly active gene product.

According to the passage, the i2 isoform of the *TERT* gene contains a portion of intron 2 and is believed to downregulate the active isoform. The passage also states that telomerase activity in somatic cells is associated with tumorigenesis, or the formation of cancerous tumors. Therefore, i2-mediated downregulation of telomerase activity in somatic cells would be expected to **reduce the risk of cancer development**.

(Choices A, C, and D) The passage lists hypertension, Alzheimer disease, and liver disease as conditions associated with decreased telomere length and senescence. Downregulation of telomerase activity by the i2 isoform of TERT would lead to shortened telomeres and senescent cells, and therefore would *increase* the risk of these conditions.

Educational objective:
DNA is initially transcribed as precursor mRNA (pre-mRNA), consisting of introns and exons. The pre-mRNA can then be spliced into one of several possible isoforms, each containing multiple exons and some containing portions of introns. Many inactive isoforms are believed to help regulate the activity of active isoforms.

Time spent:0 sec
Subject:
Biology

QID:403711
System:
Molecular Biology

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